



UMA MEMORY BUS CONTENTION & THE PRE-INFERENCE INGESTION TAX

How Unthrottled Sensor DMA Writes Starve Neural Weight Streaming on Shared LPDDR5 Memory Architectures

SENSOR INGESTION

Dual MIPI CSI-2 Cameras

Dual 1080p @ 60 FPS RGB-D

Raw Pixel DMA Bursting

Traffic: 1.2 GB/s Writes

Continuous DRAM Pushes

DMA Writes

NEURAL ACCELERATOR

NPU / Vision Encoder

ViT / DINOv2 Layer Streaming

Weight & Activation Fetches

Burst: 15–25 GB/s Reads

Systolic Array Memory Bandwidth

Weight Streaming

REAL-TIME CONTROL

1 kHz Control Thread

State Estimation & CBF Check

L3 Cache Sensitive Loop

Low Bandwidth, Hard Latency

Vulnerable to Cache Invalidation

Cache Lines



SHARED LPDDR5 SYSTEM BUS & CROSSBAR INTERCONNECT

Contention Bottleneck: Total Bandwidth Capacity = 34 GB/s · Arbiter Policy: Round-Robin vs. Priority QoS

✗ NAIVE BUFFERING (High Tax & Latency Spikes)

- User-space `memcpy()` duplicates camera frames across pages
- Camera DMA evicts CPU L2/L3 cache lines used by 1 kHz control
- NPU weight fetches stall behind unmanaged video packet bursts
- Inference Tail Spikes: $P_{50} = 22 \text{ ms} \rightarrow P_{99} = 135 \text{ ms}$ (Bus Starvation)

✓ ZERO-COPY DMA RING BUFFER (OBS-01 Architecture)

- Pre-allocated contiguous physical DRAM pool (`dma_alloc_coherent`)
- Camera writes directly into NPU-accessible virtual memory pointers
- Hardware QoS arbiter gives NPU weight reads priority over DMA lines
- Deterministic Ingestion: $t_{\text{ingest}} \leq 8 \text{ ms}$ with bounded jitter ($< 1 \text{ ms}$)